

HDOC - Day 23

Menu-driven Basic Square/Rectangular Patterns

Algorithm

1. Start and display the menu for four square/rectangular patterns and Exit.
2. Read the user choice and use switch-case to call the corresponding parameterless void function.
3. Inside the selected function, read the required rows/columns.
4. Use nested loops to print the selected pattern without using arrays.
5. Return to the menu after printing; stop when Exit is selected.

Test Case Table

Test	Operation	Input	Expected output
1	Solid square	n = 4	4 x 4 solid square
2	Solid rectangle	m = 3, n = 5	3 x 5 solid rectangle
3	Row-number pattern	m = 4, n = 4	Rows contain 1s, 2s, 3s, 4s
4	Column-number pattern	m = 4, n = 4	Each row: 1 2 3 4

C Program

```
#include <stdio.h>
void square(){int n,i,j; printf("Enter n: "); scanf("%d",&n); for(i=1;i<=n;i++){for(j=1;j<=n;j++) printf("* "); printf("\n");}
void rectangle(){int m,n,i,j; printf("Enter rows and columns: "); scanf("%d%d",&m,&n); for(i=1;i<=m;i++){for(j=1;j<=n;j++) printf("* "); printf("\n");}
void rowNumber(){int m,n,i,j; printf("Enter rows and columns: "); scanf("%d%d",&m,&n); for(i=1;i<=m;i++){for(j=1;j<=n;j++) printf("%d ",j); printf("\n");}
void columnNumber(){int m,n,i,j; printf("Enter rows and columns: "); scanf("%d%d",&m,&n); for(i=1;i<=m;i++){for(j=1;j<=n;j++) printf("%d",j); printf("\n");}
int main(){int ch; do{printf("\n1. Solid square\n2. Solid rectangle\n3. Row-number pattern\n4. Column-number pattern\n5. Exit\n");
```

Execution

Test Case 1 - Solid Square

```
puppie@Anvikas-MacBook-Air Desktop % clang hdoc23.c -o hdoc23
puppie@Anvikas-MacBook-Air Desktop % ./hdoc23
1. Solid square
2. Solid rectangle
3. Row-number pattern
4. Column-number pattern
5. Exit
Enter choice: 1
Enter n: 4
* * * *
* * * *
* * * *
* * * *
puppie@Anvikas-MacBook-Air Desktop %
```

Test Case 2 - Solid Rectangle

```
puppie@Anvikas-MacBook-Air Desktop % ./hdoc23
Enter choice: 2
Enter rows and columns: 3 5
* * * * *
* * * * *
* * * * *
puppie@Anvikas-MacBook-Air Desktop %
```

Test Case 3 - Row-number Pattern

```
puppie@Anvikas-MacBook-Air Desktop % ./hdoc23
Enter choice: 3
Enter rows and columns: 4 4
1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4
puppie@Anvikas-MacBook-Air Desktop %
```

Test Case 4 - Column-number Pattern

```
puppie@Anvikas-MacBook-Air Desktop % ./hdoc23
Enter choice: 4
Enter rows and columns: 4 4
1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4
puppie@Anvikas-MacBook-Air Desktop %
```

Result: The menu-driven C program using switch-case, nested loops and parameterless functions was implemented and tested successfully for all four patterns.